JOURNAL OF GEOMAGNETISM AND GEOELECTRICITY VOL. 17, NO. 3-4, 1965

Rockmagnetic Research at the Institut fur Angewandte Geophysik, Universität München

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Abstract

The first part of this paper describes the investigation of thermomagnetic effects in three basalt samples. It was found that when a sample was heated in air the ferrimagnetic mineral component changed from a homogeneous member of the Ulvöspinell-Magnetite-series into two components. One of these had the same O/*Fe*-ratio, the other one the same *Fe*/*Ti*-ratio as the original mineral. No changes took place when the sample was heated in a vacuum of 10^{-5} Torr.

The second part deals with the observation of domain patterns in natural magnetites and the changes of these patterns under the influence of external magnetic fields. With the help of a new polishing technique powder patterns could be observed on magnetite grains in a non-conducting rock matrix.

In the third part the magnetization of the Suevites-that are tuffaceous rocks formed in connection with the formation of the Ries crater-is discussed. At all locations investigated the magnetization is nearly anti-parallel to the present direction of the earth's magnetic field. The small scatter among the directions from different location is taken as an indication for simultaneous magnetization of all Suevites.

1. Introduction

This paper is a report on the progress made in different fields of rockmagnetic research at the Institut für Angewandte Geophysik, Universität München. It is a direct sequel to a paper read under the same title by Angenheister and Helbig at the session of the IAGA commission 3 at Berkeley in 1963. The results reported have been obtained by the three first mentioned authors respectively. They were presented at the symposium at Pittsburgh by the last mentioned author.

Research in rock magnetism at Munich University has been concentrated mainly in two directions: thermo-magnetic effects in natural titano-magnetites with special emphasis on exsolvation and oxidation phenomena (N. Petersen), and observation of domain structures in natural magnetites/(H. Soffel). J. Pohl investigated the magnetic properties of the Suevites.

2. Thermo-Magnetic Investigation of Basalts (N. Petersen)

While basalt samples of a few hundred cm^3 are, in general, homogeneous in their mag-

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netic properties,variations by a factor of ten or more have been observed in the magnetization of individual basalt flows over vertical distances of some meters (C. Turkowsky, unpublished thesis at Munich University). As these differences might be due to differences in the thermal history of different parts of the flow-different rates of cooling, differences in the amount of oxygen available- the process of exsolvation under different conditions was investigated in some detail.

Three basalt samples were used for the experiments: two came from the Vogelsberg, the third from the Rauher Kulm (Northern Bavaria), all three tertiary in age. Each sample was first measured in the natural state. Then a part of the sample was heated in air with measurements taken after 30 minutes, 100 minutes and 13 hours 40 minutes. For control, another part of the same sample underwent the same heating cycle, but in a vacuum of 10^{-5} Torr. Before the beginning of the heating cycle and after each heating period, the following observations wer made: opaque microscopy, chemical composition (*X*-ray micro-analyser), saturation magnetization vs. temperature curves (Forrer balance), hysteresis loop (in a field of 500 Oersted which was not always sufficient for saturation), saturation remanence and coercive force (after magnetization in a field of 7000 Oersted), natural remanence and thermo-remanence.

It was found that the ferrimagnetic minerals in the untreated samples belonged to the FeO-Fe_2O_3-TiO_2 system. For the sample " Rauher Kulm " the ferrimagnetic component was a member of the Ulvöspinell-Magnetite series, that of the other two samples did not differ much from this series (position just right of the Ulvöspinell-Magnetite line in the ternary system).

For the samples heated in vacuum the magnetic properties-and thus by inference the composition of the ferrimagnetic mineral component-showed practically no change. The ferrimagnetic component of the samples heated in air, however, had exsolved into a new ferrimagnetic component (Titano-Maghemite) and a non-ferrimagnetic component. The unexsolved mineral and the ferrimagnetic exsolution product had the same O/Fe-ratio (they lie on one exsolution line), while the non-ferrimagnetic exsolution product had the same Fe/Ti-ratio as the unexsolved mineral component (they lie on one oxidation line). The relation of the different mineral components is shown in figure 1.

From these observations it is concluded that the titanomaghemites ore not oxidation products, but proper exsolution products of the titano-magnetites (O/Fe-ratio remains unchanged). However, such an exsolution can only take place if concurrently some portion of the original material is oxidized. This could explain why no changes take place in a sample heated in vacuum. Moreover, these conclusions agree with the assumption, that differences in the magnetic properties of different parts from one and the same basalt flow can be caused by a different chemical environment during the cooling period: portions close to the surface of the flow are in contact with oxygene, allowing the titano-magnetites to exsolve, while in portions deeper down the partial absence of oxygene inhibits exsolution.

Further investigations of the exsolution phenomenon are planned. Complete results and a more detailed discussion will be published in the near future.

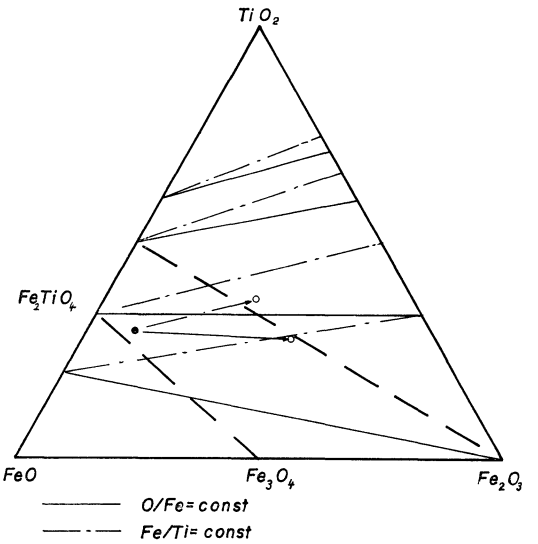


Fig. 1 Original ferrimagnetic mineral component (full circle) andend products after prolonged heating in air (open circles). The ferrimagnetic end product is formed by exsolution (O/Fe =const), the non-ferrimagnetic by oxidation (Fe/Ti=const).

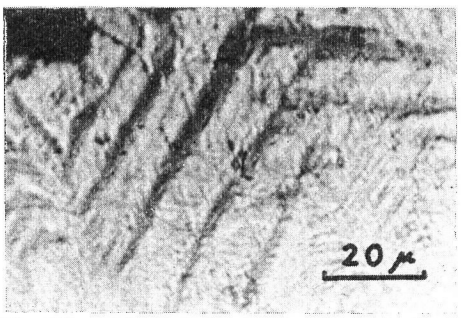
3. Observation of Domain Structures (H. Soffel).

At the IUGG-meeting at Berkeley in 1963, observations of domain structures of magnetite using Bitter's technique had been reported. As one can only observe the domain structures on polished and strain-free surfaces, the samples were polished electrolytically. Polishing by electrolytical means is restricted to samples of good electric conductivity such as magnetite single crystals and very pure and compact magnetite ores.

Investigations of domain structures can be extended to samples of low conductivity by means of a newly developed polishing method. Hereby, the specimens are heated in a vacuum better than 10^{-5}mmHg up to more than 60 hours. The surfaces thus obtained are virtually strain-free and sufficiently even. As the rate of evaporation shows an anisotropy which runs parallel with etching anisotropy, the surfaces of adjacent crystals can be at different levels after polishing. Experiments by N. Petersen have shown, that in the gross the magnetic properties (I_S and T_C) of magnetite and titanomagnetite samples are uneffected by heating in a vacuum better than 10^{-5}mmHg. No evidence, however, is available as to which extent the original domain structure is altered by the so-called thermo-polishing.

The accompanying figures are examples for different types of samples, which show a variety of domain structures. Fig. 2[1] shows powder patterns on the 111-plane of a magnetite single crystal (3mm in diameter; serpentine, Pfitscher Joch, Austria) after a thermo-polishing of two hours. Fig. 3 shows another type of domains on the same plane of the same

[1] Taken from H. Soffel, Observation and interpretation of magnetic domains in natural magnetite. Publication in preparation at Elsevier's Publishing Company, Amsterdam.



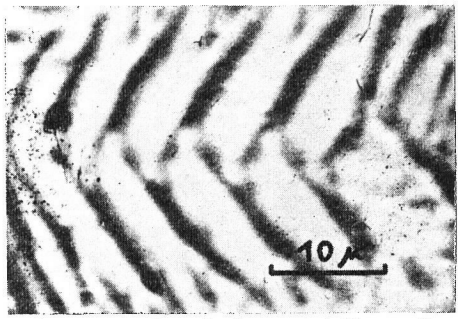
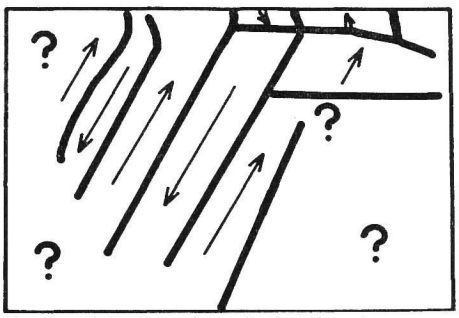


Fig. 2[1] and 3 Domain structures on the 111-plane of a magnetite single crystal (Serpentine, Pfitscher Joch, Austria) after a two hours thermo-polishing.



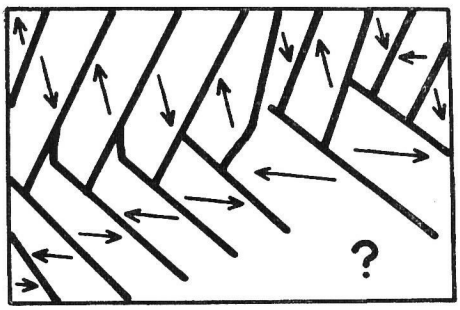


Fig. 4 and 5 Drawing of the domain structures of figures 2[1] and 3. The thin arrows indicate the direction of magnetization of the domains.

crystal. Figures 4 and 5 give the directions of magnetization within the individual domains. The directions of magnetization (especially those shown in figure 5) seem to contradict the rule, that the normal component of magnetization must be continuous across the domain boundary. The explanation for this apparent disagreement with theory lies in the fact, the 111-plane does not contain a direction of easy magnetization. Therefore, the domains are actually closure domains of the type shown in figure 6, while the main domains are mag-

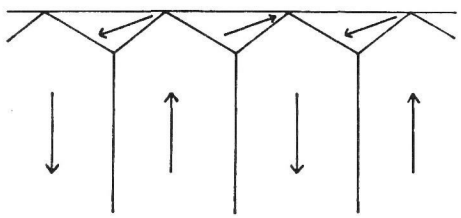


Fig. 6 Closure domains, when the axis of easy magnetization is perpendicular to the surface.

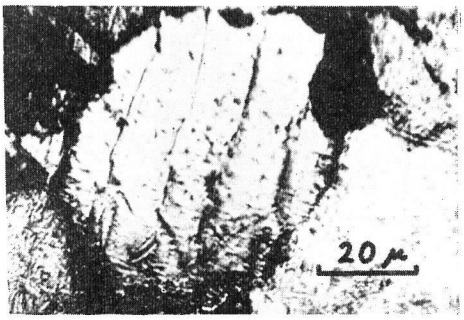


Fig. 7[2] Domain structure on a grain of a magnetite ore sample from Kirunavaara/Sweden after 18 hours of thermo-polishing.

[2] Taken from H. Soffel, Die Sichtbarmachung von Grenzen Weiss'scher Bezirke von polykristallinem Magnetit mit der Methode der Bitterschen Streifen. Zeitschrift für Geophysik 30 (1964) pp. 45–47

netized perpendicularly to the surface. All three different wall types of magnetite occur: 180°–, 109°– and 71°–walls.

Fig. 7[2] shows the domain structure observed on a grain in a polycrystalline magnetite ore (Kirunavaara, Sweden) after eighteen hours of thermo-polishing. The light colored surface is a 110–plane. The darker colored grains are also magnetite, but due to their different orientation they have been eroded more strongly. The domains visible in the figure are major domains with their magnetization parallel to the surface. Only 180°–walls can be observed. For a magnetization parallel to the surface no closure domains are to be expected.

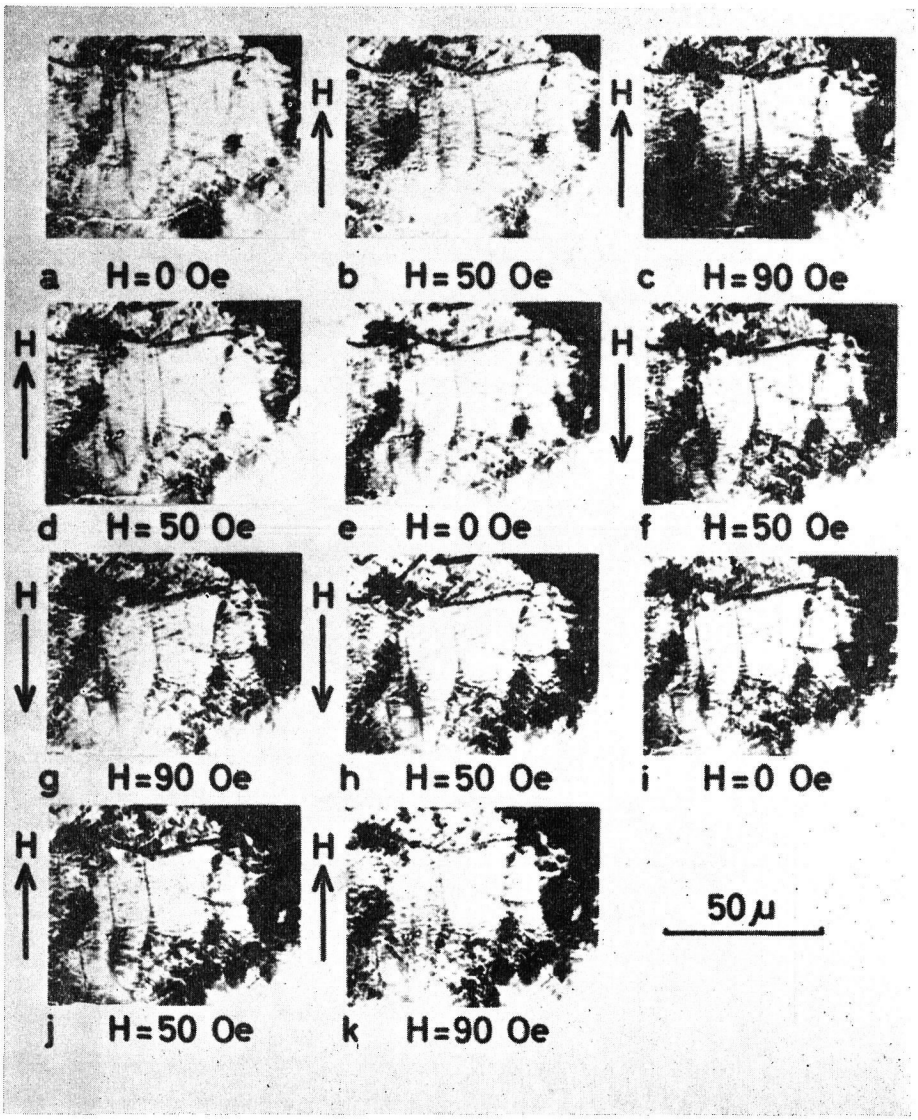


Fig. 8 Displacement of the domain boundaries of a magnetite grain under the influence of an external field of varying strength and polarity. The magnetite ore sample from Kirunavaara/Sweden was thermo-polished for 18 hours after mechanical polishing.

The observations agree with this.

Fig. 8 shows a sequence of observations of a different grain from the sample shown in figure. 7. During the observations a magnetic field of varying strength and polarity was applied parallel to the left hand margin of the frame as indicated under the individual frames. When an external field is applied, domains with a direction of magnetization parallel to the external field grow on the expense of those with directions antiparallel to the field. Though the maximal field applied in the experiment (90 Oersted) was not sufficient to saturate the sample (see frames c, g, k), an attempt was made to obtain a hysteresis loop by integrating the areas of the domains of equal direction for each frame. The resulting graph is shown in figure 9[3]. In judging the result one should bear in mind, that each point is obtained by integrating the surface of less than eight domains, and that for a proper evaluation of the

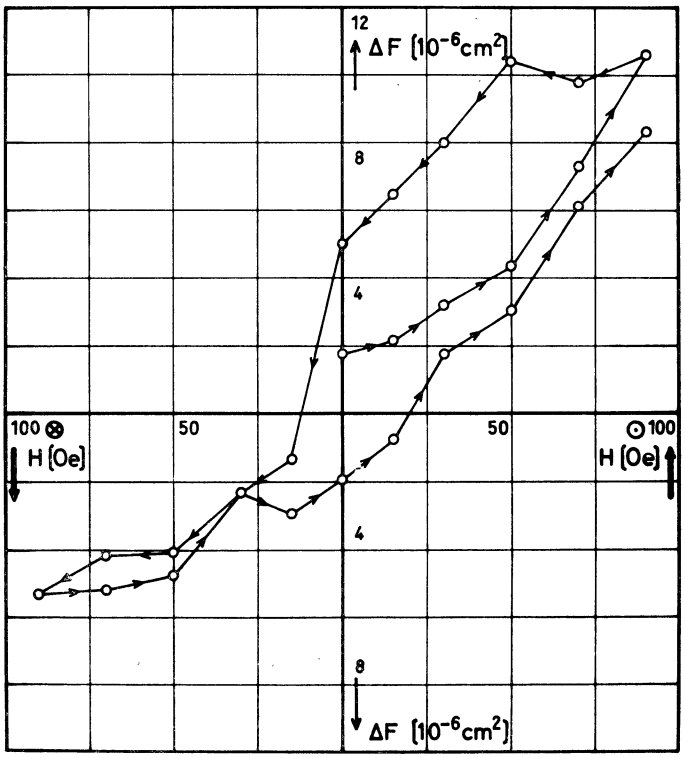


Fig. 9[3] Hysteresis loop of the experiment illustrated in figure 8.

hysteresis loop one would have to compare the volumes of the domains instead of their surface areas.

The domain structures discussed in the previous paragraphs all showed minimization of the magnetic flux across the free surface of the grain-either by closure domains or by magnetization parallel to the free surface. This distribution of magnetization is, however, by

[3] Taken from H. Soffel, Magnetic domains of polycrystalline natural magnetite. Publication in preparation in the Zeitschrift für Geophysik (1965).

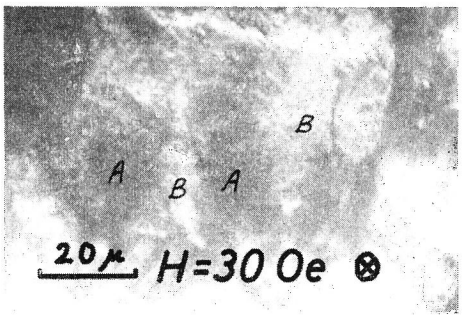


Fig. 10

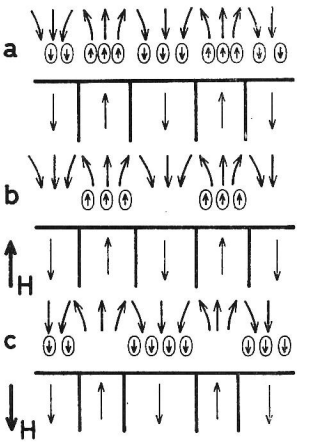


Fig. 11

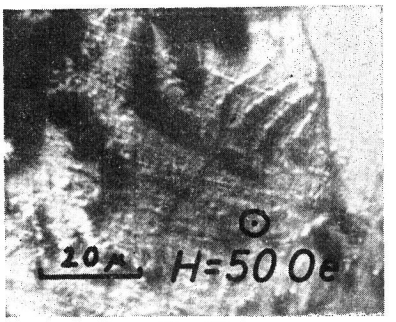
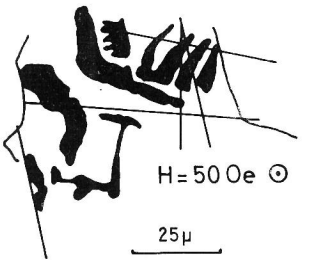


Fig. 12



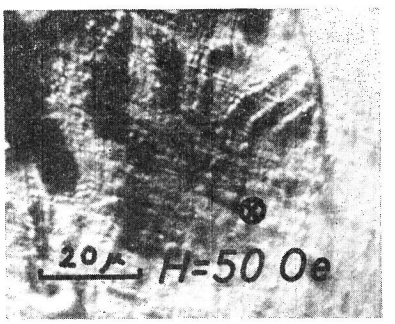


Fig. 13

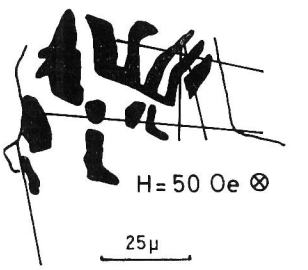


Fig. 14

Fig. 10 Domain structures on the surface of a grain in a magnetite ore from Pantano el Pintado/ Spain. The specimen was thermo-polished for 54 hours. An external field of 30 Oersted is at right angles to the surface away from the viewer.

Fig. 11 Schematic drawing of the changes of domain structures under the effect of changing polarity of the external field, when the major domains are magnetized vertically to the surface and no closure domains occur.

Fig. 12 and 13 (Same caption) Domain structures on the surface of a magnetite grain embedded in a serpentinite matrix (Quarry Wurlitz-Woja, northern Bavaria). Change of the domain structures under the effect of an external field at right angles to the surface. In fig. 12 towards the viewer, in fig. 13 away from the viewer.

Fig. 14 Drawing of the change of the domain structures of figures 12 and 13, the upper part of the frame corresponding to figure 12, the lower one to figure 13.

no means the general rule with natural magnetites. Several samples were found, where the direction of magnetization was clearly perpendicular to the surface, for about half of the surface towards the viewer, and for the remainder half away from the viewer. One such sample was a coarsly crystalline ore (Pantano el Pintado, Spain) which had been thermo-polished for 54 hours. Fig. 10 shows the domains on a crystal of this sample. The structures became visible only after a field of 30 Oersted was applied perpendicularly to the surface (in this case away from the viewer). The dark areas (A) indicate domains with magnetization parallel to the external field, the light areas (B) domains with opposite direction of magnetization as indicated in figure 11.

Another example was a magnetite grain in a serpentinite matrix (Quarry Wurlitz-Woja, northern Bavaria). Figure 12 shows a part of this grain after 33 hours of thermo-polishing (this is an example of the polishing of a magnetite grain in an electrically non-conducting matrix). Like in the case of figure 10, the structures became visible only after a field of 50 Oersted had been applied vertically to the surface (directed towards the viewer). The dark areas indicate domains with magnetization parallel to the external field, the light areas domains with antiparallel magnetization. When the external field was reversed, the distribution of the magetite colloid changed accordingly (figure 13). It is, of course, not to be expected, that the originally light areas become dark on the reversal of the external field, as at the same time the relative size of the two groups of domains changes, but it is evident, that many areas which have been light in figure 12 are dark in figure 13. In order to simplify the comparison the main content of figures 12 and 13 have been redrawn in figure 14. The thin lines are scratches on the surface of the grain and should provide a grid against which to observe the displacement of the dark areas.

Conclusions

The method of thermo-polishing allows the observation of powder patterns on surfaces of magnetite samples which cannot be polished electrolytically, such as polycrystalline ores and grains in a non-conducting rock matrix. Changes in the details of the domain structures during the heating process cannot be excluded, but the same is true with respect to the stresses during sampling, cutting and mechanical polishing.

4. The Magnetization of the Suevites in the Vicinity of the Nordlinger Ries (J. Pohl)

The Nördlinger Ries is a Krypto-volcanic structure of about twenty km diameter in southern Germany, whose origin is still controversial. Associated with the Ries crater is a negative anomaly of about 250 γ magnitude. The magnetized body causing this anomaly has not yet been identified. Such an identification would perhaps contribute to the solution of the problem whether the Ries is of volcanic or meteoritic origin.

The Suevites are tuffaceous rocks consiting of a mixture of crystalline and sedimentary material. A vitreous component, which consists mainly of molten crystalline material, contains high pressure minerals such as Coesite and Stishovite. The Suevites are found on the crater wall and outside the crater in the vicinity of the wall. The purpose of the present

study was to determine whether magnitude and direction of the magnetization of the Suevites would agree with the assumption that the negative anomly within the crater could be caused by Suevite masses not yet located.

Oriented samples were taken from all presently known locations (figure 15). Cubes were cut from the samples and the magnetization was measured with a fluxgate magnetometer. For all locations the magnetization was nearly antiparallel to the present geomagnetic field (table 1). The mean natural remanent magnetization was 59.10^{-5} Γ. The inclinations varied from location to location between $-25°$ and $-61°$, the easterly declination between 141° and 202°. The magnetization observed is such that at least part of the negative anomalies in the Ries could be attributed to Suevites.

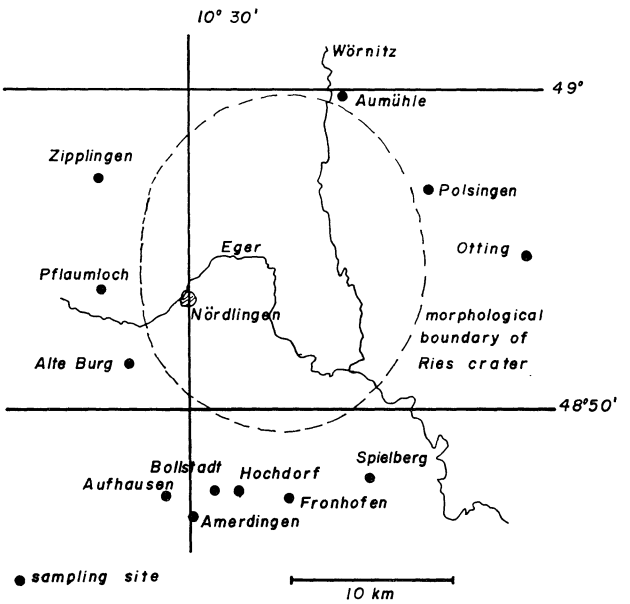


Fig. 15 Location of the Ries crater and sampling sites. The magnetization of the Suevites at all sampling sites is practically antiparallel to the present magnetic field. The directions at different locations are closely parallel to each other.

Most of the samples were remeasured after a. c.-demagnetization. For field ampitudes smaller than 200 Oersted no changes in the magnetization of the samples were observed, but with field amplitudes up to 1000 Oersted the magnitude of the magnetization was in most cases reduced to a few percent of the original value. After the demagnetization the scatter in the direction of the magnetization at different sites was strongly reduced (see table 1). The mean inclination after demagnetization was $-60°$, the mean declination 191°, and the radius of the 95 percent confidence circle from all locations 1.5°.

The small scatter in the directions determined at different locations indicates that all Suevites which have been investigated obtained the magnetization simultaneously, presumably at or immediately after the Ries event. Geological determinations place this event in

the Miocene at the turn from Tortonian to Sarmatian, Potassium-Argon dating of the vitreous component of the Suevites yields an age of 14.8±7 million years.

Table 1

Site	n	$\bar{J}_{nr}$	$\bar{Q}$	$\bar{i}$	$\bar{\delta}$	θ
Zipplingen	10 6	20	1.3	−25° −60°	141° 186°	33° 10°
Pfiaumloch	7 5	30	1.7	−54° −60°	195° 189°	7° 8°
Altebürg	10 7	31	3.1	−53° −59°	202° 197°	11° 6.5°
Aufhausen	28 18	59	3.9	−55° −58°	192° 192°	3° 1.5°
Amerdingen	27 17	92	4.9	−57° −58°	194° 193°	2° 2°
Bollstadt	32 19	137	5.7	−60° −60°	190° 189°	2.5° 1.5°
Hochdorf	14 6	59	4.5	−57° −60°	202° 195°	2.5° 4.5°
Fronhofen	5 5	127	30	−61° −61°	194° 195°	2° 3°
Spielberg	12 6	81	6.5	−57° −61°	197° 194°	2° 2°
Otting	18 14	56	4.9	−58° −60°	191° 190°	4° 3.5°
Polsingen	7 4	13	9	−56° −62°	187° 186°	5.5° 6.5°
Aumühle	9 4	4	1.5	−56° −55°	193° 190°	3° 7°

n number of samples
$\bar{J}_{nr}$ mean natural remanent magnetization
$\bar{Q}$ mean Königsberger-ratio
$\bar{i}$ mean inclination
$\bar{\delta}$ mean declination
θ 95 percent radius of confidence

The second line for each site refers to measurements after a. c.-demagnetization.